Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/03/2025
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Effect of Head-Down Tilt +/- CO2 on Human Glymphatic Function		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-8205	**Congressional District:**	3
Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC020N0001-FLAGSHIP, OMNIBUS1 Human Research Program: Crew Health Appendix A; Omnibus1-Appendix B
Start Date:	01/06/2023	**End Date:**	06/30/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2026 per V. Lehman/JSC and NSSC information (Ed., 4/30/25) NOTE: End date changed to 06/30/2025 per V. Lehman/JSC (Ed., 3/18/25) NOTE: End date changed to 12/31/2024 per NSSC information (Ed., 7/23/24) NOTE: End date changed to 08/31/2024 per M. Stenger/NASA HHC Element Scientist and C. Ribeiro/NASA HHC (Ed., 1/12/24)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Richmond, Sutton Ph.D. (University of Florida, Gainesville) Roy, Arkaprava Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC23K0365		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight Associated Neuro-ocular Syndrome (SANS) is a potentially mission-limiting condition for astronauts, yet its underlying pathophysiology is poorly understood. It has been proposed that impairment of the glymphatic system in response to fluid shift or venous congestion could be an underlying cause of SANS. However, there have been no studies to date evaluating human glymphatic function in response to simulated microgravity. While our team is currently evaluating the effect of elevated carbon dioxide (CO2) on human glymphatic function (awards from NASA and the Office of Naval Research), we have yet to evaluate the spaceflight-relevant combination of CO2 and simulated microgravity. (Ed. Note: See related Grant #80NSSC17K0021--PI: Seidler). The latter can be achieved via 6° head-down tilt (HDT), simulating the headward fluid shifts of microgravity. Here, we propose to conduct glymphatic magnetic resonance imaging (MRI) in human subjects over a span of several hours while they are either supine or lying on a 6° foam wedge with head-down tilt. They will also be breathing elevated CO2 to better mimic the conditions on the International Space Station. We will measure the effects of HDT positioning on human glymphatic function (Aim 1) and the combined effects of breathing elevated CO2 and HDT on human glymphatic function (Aim 2). Thus, we will quantify the acute effects HDT with and without elevated CO2.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is examining the effects of elevated CO2 combined with and without head down tilt on the brain's waste clearance system (glymphatic system). Understanding the impact that elevated CO2 has on this process may inform environmental health.
Task Progress:	In the proposed study, we will investigate the individual and combined effects of simulated microgravity and hypercapnia on human glymphatic function. Using acute head-down tilt (HDT) bedrest and exposure to elevated CO2, this study will be a critical first step toward understanding the potential interactions of microgravity and elevated CO2 on perivascular glymphatic function and brain health. SPECIFIC AIM 1: To measure the effects of HDT positioning on human glymphatic function. SPECIFIC AIM 2: To assess the combined effect of breathing elevated CO2 and HDT on human glymphatic function. We have completed data collection with five participants and another three are in progress. The preliminary findings were presented at the NASA Human Research Program (HRP) Investigators' Workshop (IWS) in Galveston, TX. Our previous project in which we were testing the effects of elevated CO2 in supine participants revealed that the presence of elevated CO2 slowed glymphatic clearance from the vitreous chamber of the eye. That is, there was greater contrast-induced signal intensity on the gas day relative to the ambient air day at the 90-minute post-contrast injection time point. With the preliminary findings that we presented at the NASA HRP IWS, the results suggest that the addition of the head-down tilt positioning slows glymphatic clearance from this region at this 90-minute time point even more than elevated CO2 alone. It remains to be seen if this finding holds with the collection and analysis of additional participants' data and statistical assessment.
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Abstracts for Journals and Proceedings	Seidler RD, Odor R, Richmond SB, Rao R, Rane S, Kernagis DN, Albayram MS, Rosenberg JT, Iliff JJ, Peng TJ. "Effects of head-down tilt +/- CO2 on human glymphatic function." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025